

$$x^2 + 26x = 27$$

Step 1 - write the equation

Step 2 - Build, but halve 26x

Whiteboard showing the equation $x^2 + 26x = 27$. Below it, algebra tiles are arranged: a blue tile for x^2 , a green tile for $26x$, and a yellow tile for 27 .

Whiteboard showing the equation $x^2 + 26x = 27$. Below it, algebra tiles are arranged in a square. The top row has a blue tile for x^2 and a green tile for $13x$. The bottom row has a green tile for $13x$ and a pink tile for 169 . A bracket on the left indicates the side length is $x+13$, and a bracket on top indicates the side length is $x+13$. The right side is a yellow tile for 27 .

Step 3 → We "complete the square"

Whiteboard showing the equation $x^2 + 26x = 27$. Below it, algebra tiles are arranged in a square. The top row has a blue tile for x^2 and a green tile for $13x$. The bottom row has a green tile for $13x$ and a pink tile for 169 . A bracket on the left indicates the side length is $x+13$, and a bracket on top indicates the side length is $x+13$. The right side is a yellow tile for 27 and a pink tile for 169 , with a plus sign between them.

Whiteboard showing the equation $x^2 + 26x = 27$. Below it, algebra tiles are arranged in a square. The top row has a blue tile for x^2 and a green tile for $13x$. The bottom row has a green tile for $13x$ and a pink tile for 169 . A bracket on the left indicates the side length is $x+13$, and a bracket on top indicates the side length is $x+13$. The right side is a yellow tile for 196 .

Step 4

Work out side lengths by taking square root.

$x^2 + 26x = 27$

$x+13$

x^2 $13x$

$13x$ 169

$x+13$

196

13×14

So $x+13=14$
 $x=1$

note: During this time, the negative solution wasn't considered.

For thousands of years, mathematicians were oblivious to the negative solutions to their equations because they were working in the real world.

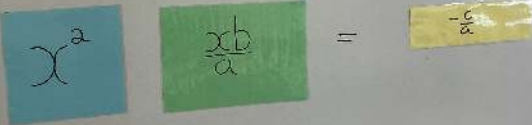
$$(x+13)^2 = 196$$

$$x+13 = \pm 14$$

$$x = 1 \text{ or } 27$$

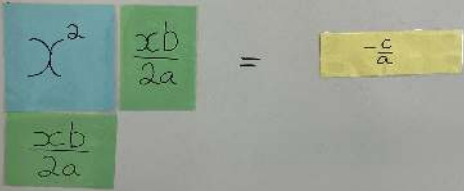
Step 5

↳ Proof of quad formula

$$\begin{aligned} ax^2 + bx + c &= 0 \\ ax^2 + bx &= -c \\ x^2 + \frac{b}{a}x &= \frac{-c}{a} \end{aligned}$$


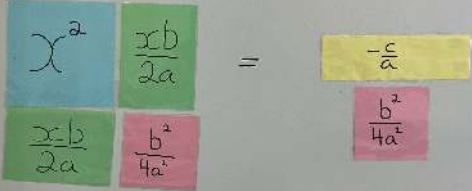
} the setup.

Step 6

$$\begin{aligned} ax^2 + bx + c &= 0 \\ ax^2 + bx &= -c \\ x^2 + \frac{b}{a}x &= \frac{-c}{a} \end{aligned}$$


Step 7

complete the square

$$\begin{aligned} ax^2 + bx + c &= 0 \\ ax^2 + bx &= -c \\ x^2 + \frac{b}{a}x &= \frac{-c}{a} \end{aligned}$$


Step 8

Build $\rightarrow$ algebra to proof.

$$\begin{aligned} ax^2 + bx + c &= 0 \\ ax^2 + bx &= -c \\ x^2 + \frac{b}{a}x &= \frac{-c}{a} \end{aligned}$$
$$\begin{array}{|c|c|} \hline x^2 & \frac{xb}{2a} \\ \hline \frac{xb}{2a} & \frac{b^2}{4a^2} \\ \hline \end{array} = \frac{b^2 - 4ac}{4a^2}$$

$\frac{b^2}{4a^2} - \frac{4ca}{4a^2}$

$$\begin{aligned} ax^2 + bx + c &= 0 \\ ax^2 + bx &= -c \\ x^2 + \frac{b}{a}x &= \frac{-c}{a} \end{aligned}$$
$$\begin{array}{|c|c|} \hline x^2 & \frac{xb}{2a} \\ \hline \frac{xb}{2a} & \frac{b^2}{4a^2} \\ \hline \end{array} = \frac{b^2 - 4ac}{4a^2}$$
$$\left(x + \frac{xb}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\frac{b^2}{4a^2} - \frac{4ca}{4a^2}$